

Practical 3

3. Analysis of Windows registry

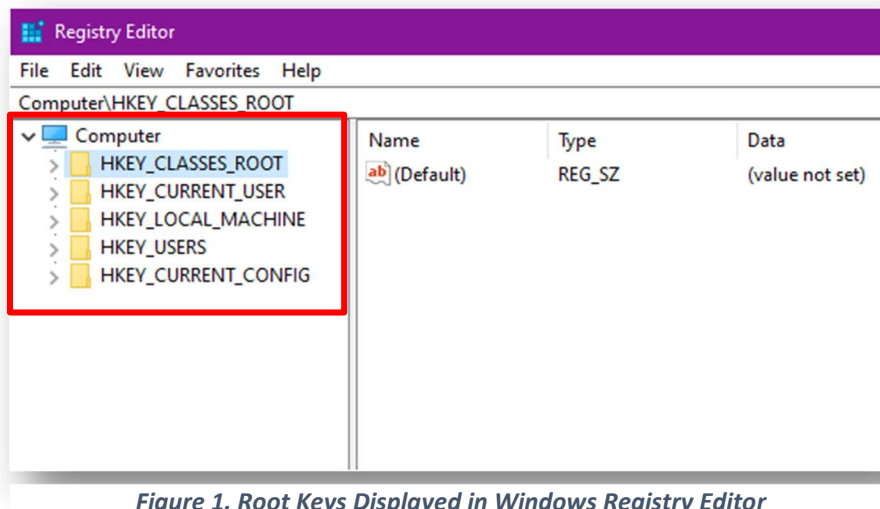
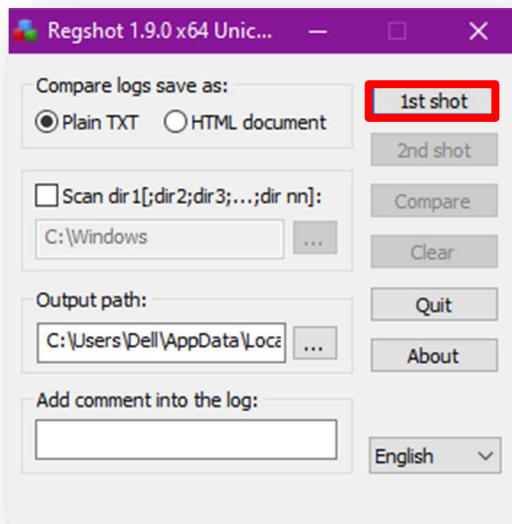


Figure 1. Root Keys Displayed in Windows Registry Editor

- ❖ The Windows Registry is a **central hierarchical database** that stores:
 - a) System configuration
 - b) Hardware & driver info
 - c) Installed applications
 - d) User activities
 - e) User-specific settings (MRU, Run commands, USB devices etc.)
- ❖ It is extremely important for forensic investigations because it contains **user behaviour traces**.
- ❖ The **Figure 1** shows the Windows Registry Editor displaying the five main root keys of the registry.
- ❖ These top-level hives—
 - a) **HKEY_CLASSES_ROOT**,
 - b) **HKEY_CURRENT_USER**,
 - c) **HKEY_LOCAL_MACHINE**,
 - d) **HKEY_USERS**, and
 - e) **HKEY_CURRENT_CONFIG**
- ❖ They organize all system, user, and application settings in Windows. They form the starting point for navigating the registry during analysis.
- ❖ Tools Used for Registry Forensics :
 - **Regshot**
 - Registry Editor (regedit) [**Figure 1**]
 - MUICache View
 - Process Monitor
 - USBDeview
 - RegRipper
- ❖ For the practical, **Regshot** is used to compare registry before and after software installation.

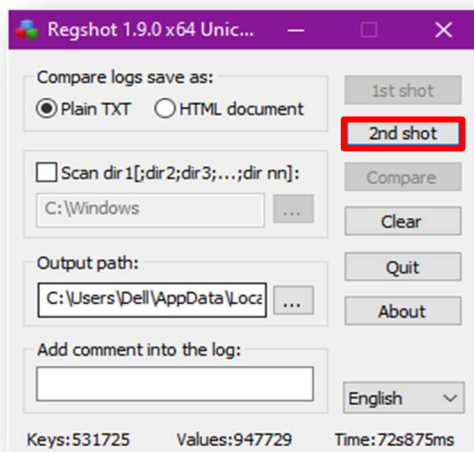
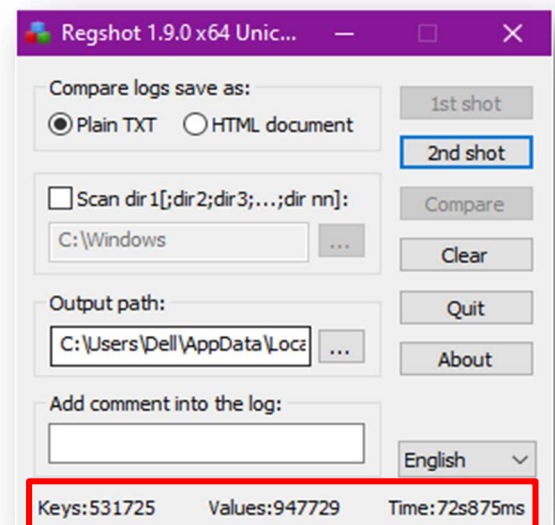
Windows Registry Analysis Using Regshot



STEP 1 — Take the First Registry Snapshot
Open **Regshot** and take the first snapshot.

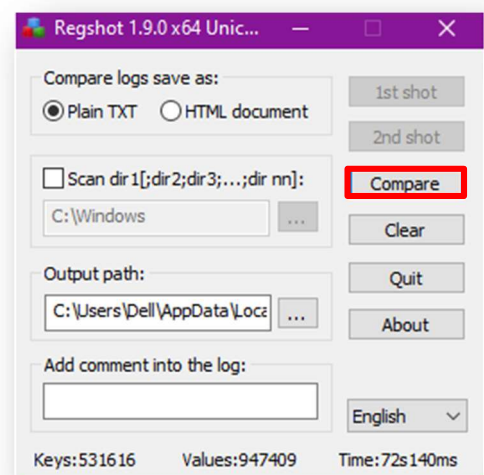
STEP 2 — Install a Software or Perform an Activity

- Now install any software or perform some action (e.g., opening a file, plugging USB, modifying settings).
- This creates **changes in Registry keys**.



STEP 3 — Take the Second Registry Snapshot
After installation, take the **second Regshot snapshot**.

STEP 4 — Compare both Snapshots
Click **Compare**.



STEP 5 — View Registry Changes

Regshot **displays :**

- **New Keys Added**
- **Values Changed**
- **Deleted Entries**

See the output through this link : [~res-x64 0000.txt](#)

Why Registry Analysis Matters ??

- ❖ **Registry artifacts help forensic investigators confirm:**
 - **Applications installed**
 - **User login activity**
 - **USB device history**
 - **Browser history traces**
 - **Last opened documents**
 - **Malware persistence keys**
- ❖ **It is a critical part of timeline analysis and intrusion reconstruction.**